Analysis of Closed-loop Inertial Gradient Dynamics

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Continuous Global Minimisation

1. In modern ML and Image processing we face continuous, single-objective unconstrained smooth minimisation problems. Such problems in the finite-dimensional setting are of the form:

$$\min_{x \in \mathbb{R}^d} f(x) \quad \text{where } f : \mathbb{R}^d \longrightarrow \mathbb{R}.$$

2. Global optimisation is not easy, as it is very easy to get stuck in saddle points and local stationary points because these methods are essentially **BLACK-BOX** or ORACLE based where we only have information at the point of inquisition [1].
3. In our study we only use first-order information of gradients of a continuous function and therefore our Oracle is of the first order. Gradient computations are relatively cheap and hence make algorithms faster.

Background & Preliminaries

- In this paper, we assume the cost function $f(x)$ is such that $f : \mathbb{R}^d \mapsto \mathbb{R}$ is a twice continuously differentiable convex function such that for all $\epsilon \in [0, 1]$ [2]:

$$f(\epsilon x + (1-\epsilon)y) \leq \epsilon f(x) + (1-\epsilon) f(y). \tag{1}$$

- We consider L-Lipschitz continuous gradients in our analysis:

$$\|\nabla f(y) - \nabla f(x)\| \leq L\|y - x\| \;\; \forall x, y \in \mathbb{R}^d. \tag{2}$$

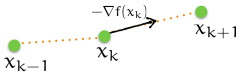
- For L-smooth convex functions, where we choose the step-size as $s = \frac{1}{L}$, $\forall\, x, y \in \mathbb{R}^d$, for $|x^*| < \infty$ we have

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2. \tag{3}$$

Nesterov Accelerated Gradient Descent

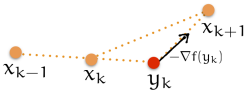
Gradient Descent

$$x_{k+1} = x_k - \epsilon \nabla f(x_k)$$



Accelerated GD

$$x_{k+1} = y_k - \epsilon \nabla f(y_k)$$
$$y_k = x_k + \frac{k-1}{k+2}(x_k - x_{k-1})$$



when $\|\nabla^2 f\| \leq \frac{1}{\epsilon}$

$$O\left(\frac{1}{\epsilon k}\right) \xrightarrow{\text{accelerated!}} O\left(\frac{1}{\epsilon k^2}\right) \quad \text{optimal rate}$$

Figure: The process of acceleration leading to faster optimisation [3].

Continuous-time Analysis

Upon using the discretisation method, which can be simplified to $t = k\sqrt{s}$, Su et al. [4], found a generalised method as an ordinary differential equation. They observed that the same ODE holds for all variations of the Nesterov's method, for asymptotic approximations.

$$\ddot{x} + \frac{3}{t}\dot{x} + \nabla f(x) = 0$$

We observe the asymptotic convergence rate of

$$f(x) - f(x^*) \leq \mathcal{O}\left(\frac{1}{t^2}\right).$$

This is used in our analysis for the general inertial gradient dynamical system [5] by defining a candidate Lyapunov function $W(t)$ for all damping functions γ such that:

$$W(t) = \frac{1}{2}\|\dot{x}\|^2 + f(x) - f^* \implies \dot{W} = -\gamma(t)||\dot{X}||^2 \leq 0 \;\; \forall \gamma \geq 0.$$

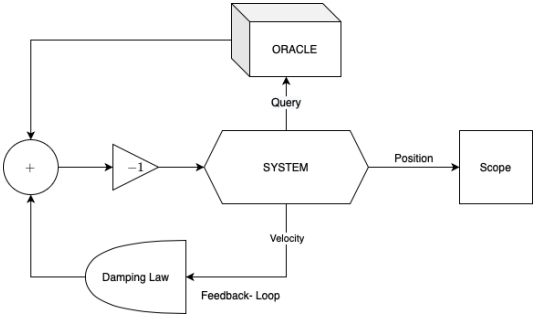
which proves global asymptotic stability.

A control system perspective

We study the the generalisation of the differential equation proposed by Su et al. [4], in real time, as a control system.

$$\ddot{x} + \gamma(t)\dot{x} + \nabla f(x) = 0.$$

We study this as a state space diagram as shown. Note that x^* is the optima in the continuous time space $\mathcal{X}^*$, the implicit solution to the ODE being $x(t)$.



The Whiplash Gradient Descent

Previously [6], we arrived at our hybrid gradient descent optimisation method, the *whiplash inertial gradient optimisation method* using a non-linear feedback constraint:

$$\ddot{X} + (1 + t\,||\dot{X}||^2\,)\dot{X} + \nabla f(X) = 0.$$

The existence and uniqueness has already been discussed for a much more general set of systems in [7]. Upon *symplectic discretization* we arrive at:

$$(\frac{1}{\sqrt{s}})\frac{v_{k+1} - v_k}{\sqrt{s}} + (1 + k\sqrt{s}||v_k||^2)\frac{v_k}{\sqrt{s}} + \nabla f(x_k) = 0.$$

Rearranging

$$v_{k+1} = (1 - \sqrt{s} - ks\; v_k^T \cdot v_k)v_k - s\nabla f(x_k)\,.$$

Unrolling this, we have, for the position after the first few steps:

$$x_{k+1} = \underbrace{x_k - s\nabla f(x_k)}_{\text{Gradient descent}} + \overbrace{\alpha_k\, z_k}^{\text{Momentum based scaling}}$$

The Whiplash Gradient Descent

Numerically 10^5 times faster than any other deterministic accelerated methods for the non-convex Rosenbrock function!

We finally have the consolidated scheme as:

$$\hat{\mathcal{W}} = \begin{cases} x_1 = x_0 - s\nabla f(x_0), \\ z_k = x_k - x_{k-1}, \\ \alpha_k = 1 - \sqrt{s} - ks||z_k||^2, \\ z_{k+1} = \alpha_k\, z_k - s\nabla f(x_k). \end{cases}$$

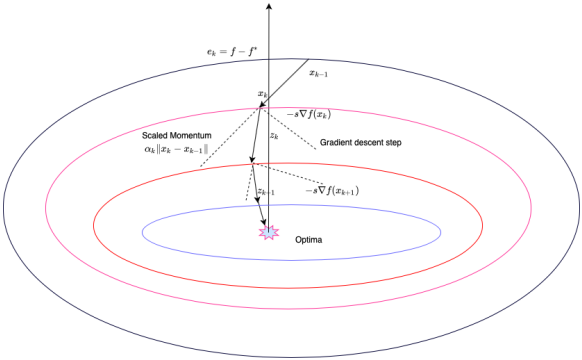


Figure: Whiplash Gradient Optimization method

Convex cost performance

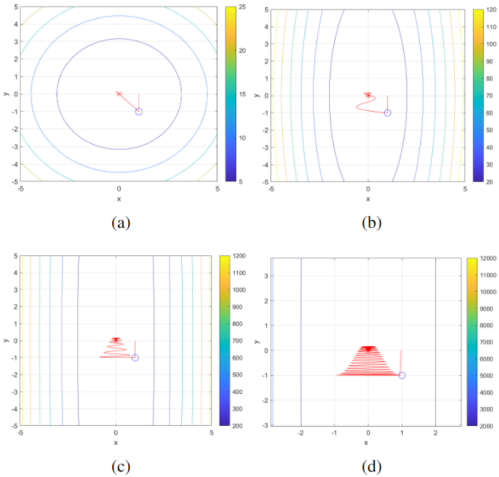


Figure: A study of performance of the whiplash gradient descent for increasing condition numbers: (a) $\kappa = 1$ (b) $\kappa = 10$ (c) $\kappa = 100$ (d) $\kappa = 1000$.

Comparative study in discrete-time

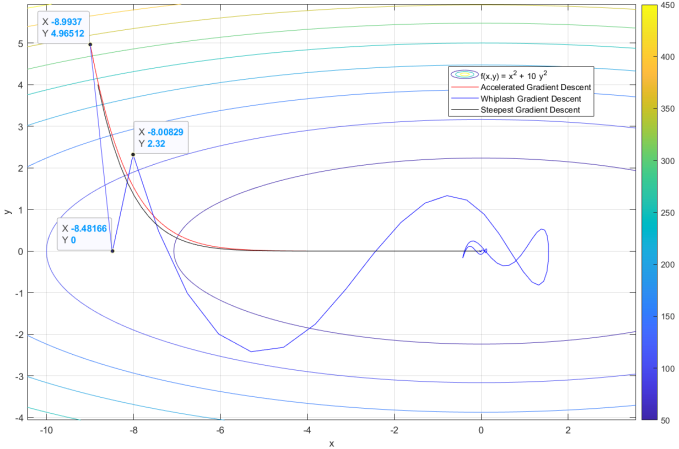


Figure: Quadratic cost function with condition number 10.

Norm Convergence & Relaxation Sequences

Theorem: Norm Convergence

The momentum sequence of the whiplash gradient descent $\|x_k - x_{k-1}\|$ converges at the rate $\mathcal{O}(\frac{1}{\sqrt[3]{k}})$ for an L-smooth convex cost function.

Theorem: Relaxation sequence for L-smooth convex condition

The whiplash gradient scheme admits a convergent relaxation sequence $\{\delta_k\}_{k=0}^{\infty}$ for convex L-smooth cost functions, so that

$$\|x_{k+1} - x^*\| \leq \|x_k - x^*\| + \sqrt{\delta_k}, \quad \forall\, k > 0,$$

where $\delta_k = \|z_k\|^2 + 2\|z_k\|\|x_k - x^*\| + \|(\frac{1}{\sqrt{L}} + \frac{k}{L}\|z_k\|^2)z_k\|^2$.

Checkpoint

What do we have:

1. Controlled Relaxation $\longrightarrow$ Controlled Oscillations $\longrightarrow$ Faster Optimization!
2. Systems are globally asymptotically stable for all damping laws which are strictly positive [6].
3. We further know that if the gradient of the cost function is Lipschitz continuous, there exists a global solution which is convergent, but such a solution is non-analytical in most cases.
4. To understand the rate of convergence of the system, we need to perform Lyapunov analysis on the Lipschitz continuous scaled system $y(t) = \nu(t)x(t)$.
5. Creating Lyapunov functions is a tedious affair and often lacks any kind of motivational rigour.

Lyapunov's fundamental argument lead to a family of energy functions which are non-increasing along the system's dynamics, which is typically the only constraint.

Asymptotic lower bound: Introduction to (Script) $\mathscr{O}$

A function $\mathscr{F}(t)$ which is asymptotically lower bounded is said to be convergent at the rate $\nu(t) > 0$, if

$$0 \leq \overline{\lim_{t\to\infty}} |\nu(t)\mathscr{F}(t)| < \infty \implies \mathscr{F}(t) = \mathscr{O}(\frac{1}{\nu(t)}). \tag{4}$$

We define the lower bound notation of two real and bounded functions $\mathcal{A}_t, \mathcal{B}_t : \mathbb{R}^+ \mapsto \mathbb{R}$, such that $\mathcal{A}_t = \mathcal{O}(a(t))$ and $\mathcal{B}_t = \mathcal{O}(b(t))$, where $\mathscr{O}(\mathrm{a}(t)) \leq \mathscr{O}(\mathrm{b}(t))$. We further have the following relations:

$$\begin{gathered}
\mathscr{O}(\lambda\,\mathrm{a}(t)) = sgn(\lambda)\,\mathscr{O}(\mathrm{a}(t)) \text{ where } \lambda \in \mathbb{R}, \\
\mathscr{O}(a(t)) \leq \lambda\mathscr{O}(b(t)) \Rightarrow \mathcal{A}_t \sim sgn(\lambda)\,\mathscr{O}(b(t)), \\
\mathscr{O}(\mathrm{a}(t)\mathrm{b}(t)) = \mathscr{O}(\mathrm{a}(t))\mathscr{O}(\mathrm{b}(t)), \\
\mathscr{O}(\mathrm{a}(t) \pm \mathrm{b}(t)) = \pm\,\mathscr{O}(\mathrm{b}(t)).
\end{gathered} \tag{5}$$

We have a combination of 2 concepts:

- Loose Upper Bound
- Strict Lower Bound

Envelope Convergence

1. In [6], we defined the converger $\mathcal{P}_t > 0$, such that for a convex cost function f, we have a convergence rate of $\mathcal{O}(\frac{1}{\mathcal{P}_t})$. This converger must be defined $\forall\, t > 0$, such that $\mathcal{P}_t \to 0$ as $t \to \infty$. A simple way of numerically verifying these results is by *enveloping* the output. We show a convergence rate for $x(t)$; i.e. we define $y(t) = \mathcal{P}_t(x(t) - x^*)$, and demonstrate its numerical convergence.
2. We consider an inertial gradient dynamical system of the form:

$$\ddot{x}(t) + (1 + \alpha(t))\dot{x}(t) + \nabla f(x(t)) = 0, \tag{6}$$

where the scalar function $\alpha(t)$ is a closed-loop control law. We check for a converger $\mathcal{P}_t$ for the system (6), such that $y(t) = \mathcal{P}_t x(t)$ is numerically convergent. We consider two types of rates: exponential (linear) and polynomial (sub-linear), characterised by $\eta > 0$ as the *strength of convergence* in continuous-time.

Envelope Convergence

Exponential Rate	Polynomial Rate
We start with $\eta = 0.1$ and keep running simulations for increasing values of η by steps of size 0.05. If a transition to the stability exists in this region, it is sharp and moves from bounded to divergent solutions, within a range of 0.01. Hence, once the transition is observed, we vary η by steps of size 0.001. If stability is observed for a critical range, we check different starting conditions.	We start with $\eta = 0.5$ and keep running simulations for increasing values of η by steps of size 0.1. If a transition to the stability exists in this region, it is sharp and moves from bounded oscillations to divergent within a range of 0.05. Hence, once the transition is observed, we vary η by steps of size 0.01. If stability is observed for a critical range, we check different starting conditions.

Once we have verified this, we can define a system such that there exists a energy function $\mathcal{E}(t)$, which for a particular integral anchor constraint, proves the convergence rate of the objective value.

Envelope Convergence

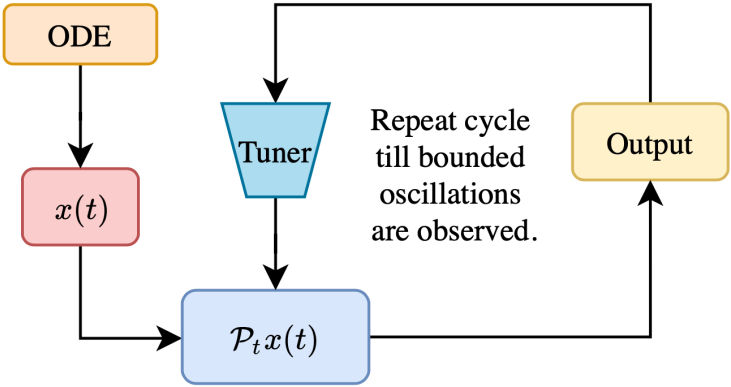


Figure: Summary of the envelope convergence method

Envelope Convergence

Theorem: Bhattacharjee-Petersen-Lyapunov Theorem

For a twice differentiable convex cost function f, the system (6) converges with a rate of:

$$f(x) - f^* = \mathcal{O}(\frac{1}{\mathcal{P}_t}), \tag{7}$$

for a converger $\mathcal{P}_t > 0$ and $\dot{\mathcal{P}}_t > 0$, such that $\mathcal{P}_t x(t)$ is bounded and convergent, if there exists a constant $c \geq 0$, for all $t > 0$ and for every solution to (6), $\mathscr{I} + c \geq 0$ where

$$\mathscr{I}_t = \int_0^t \left(\langle \alpha \mathcal{P}_s \dot{x}, x - x^* \rangle - \langle \mathcal{P}_s \nabla f(x), \dot{x} \rangle - \frac{\dot{\mathcal{P}}_s}{2} \|x - x^* + \dot{x}\|^2 \right) ds, \tag{8}$$

using the energy function for a globally asymptotically stable Lienard system, given as:

$$\mathcal{E}_t = \underbrace{\mathcal{P}_t(f(x) - f^*)}_{\text{Potential energy}} + \underbrace{\frac{\mathcal{P}_t}{2}\|x - x^* + \dot{x}\|^2}_{\text{Mixed energy}} + \underbrace{\mathscr{I}_t + c}_{\text{Stored energy}} \geq 0 \implies \dot{\mathcal{E}}_t \leq 0. \tag{9}$$

Integral anchor constraint

We can prove that:

$$\mathscr{I}_t \geq c > 0,$$

for every solution if we make a few reasonable assumptions on the ODE's GLOBALLY ASYMPTOTICALLY STABLE solutions on the basis of two case studies.

Case I: *Polynomial convergence setting*: $(t \geq t_0 > 0)$

1. $\nu(t) = t^\theta$ where $\theta > 0$.
2. $\alpha(t) = \mathscr{O}(\frac{1}{t^p})$ where $p \geq 0$.
3. Consider the polynomial converger $\mathcal{P}_t = t^\eta$. The **critical condition** is:

$$\boxed{x - x^* \sim \mathscr{O}(\frac{1}{t^\theta}) : \ \theta > \frac{\eta}{2} > 0.} \tag{10}$$

Integral anchor constraint

Case II: *Exponential convergence setting* $(t \geq t_0 \geq 0)$

1. $\nu(t) = e^{-\theta t}$ where $\theta > 0$.
2. $\alpha(t) = \mathscr{O}(e^{-pt})$ where $p \geq 0$.
3. Consider the exponential converger $\mathcal{P}_t = e^{\eta t}$ where $0 < \eta < 2\theta$. The **critical condition** is:

$$\boxed{x - x^* \sim \mathscr{O}(e^{-\theta t}) : \ \theta > \frac{\eta}{2} > 0.} \tag{11}$$

Thus, we complete the proof of Theorem 5 by proving the integral anchor constraint assumption, to include every possible solution to the dynamics (6), such that we may consider the rate of convergence for the objective $f(x) - f^*$.

Note that this theorem does not rely on a specific rate but an asymptotically bounded form only!

Verification of the envelope convergence method for the Whiplash method

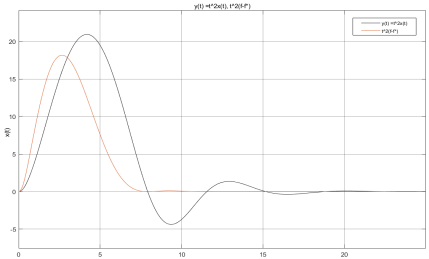


Figure: Verifying a polynomial convergence rate of $\mathcal{O}(\frac{1}{t^2})$ with the cost function $f(x) = \frac{\lambda}{2}x^2$. Experimentally verified for variations on λ.

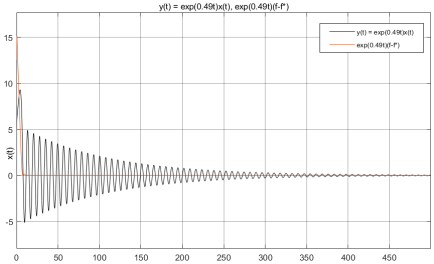


Figure: A numerical convergence rate of $\mathcal{O}(e^{-0.495t})$ is verified for the whiplash gradient descent for the cost function, $\tilde{f}(x) = \frac{1}{2}\|x\|^2$ $(\mathbb{R}^2)$.

Conclusion

1. We have reduced the analytical task of proving convergence rates by finding Lyapunov functions into the prediction of rates of convergence by finding appropriate convergers, which satisfy the design criterion.
2. Therefore, as long as we choose a target η with respect to a suitable θ, which does not violate the design criterion, our analysis holds true, which is a very strong tool for further analysis.This means *numerically* for a parameter θ, and for a range of choice on the parameter η, if

$$\boxed{\nu_t(\theta)(x(t) - x^*) \to 0 \implies f - f^* = \mathcal{O}(\frac{1}{\mathcal{P}_t(\eta)}).} \tag{12}$$

3. Our future work consists of:
   1. Stochastic gradients [8]
   2. Non-smooth analysis using Clarke sub-differential methods [9]
   3. Perturbation analysis of the system for escaping local minimas and saddle points [10]

   Looking at these perspective from a Control system Point of View!

Thank you for your kind attention!

Any Questions or Comments?

Please find the GITHUB repository link to the code and models here.

References

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